

Construction Site Robotic Arm

GROUP C

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CONTENTS

- INTRODUCTION
- BLOCK DIAGRAM
- CIRCUIT DIAGRAM
- CONSTRUCTION
- TESTING
- RESULTS
- LESSON LEARNT
- PROJECT MANAGEMNET
- REFERENCES

INTRODUCTION

- The project picks and places material in the construction site.
- The distance between the ultrasonic sensor and the detected material will be monitored on the Blynk GUI.

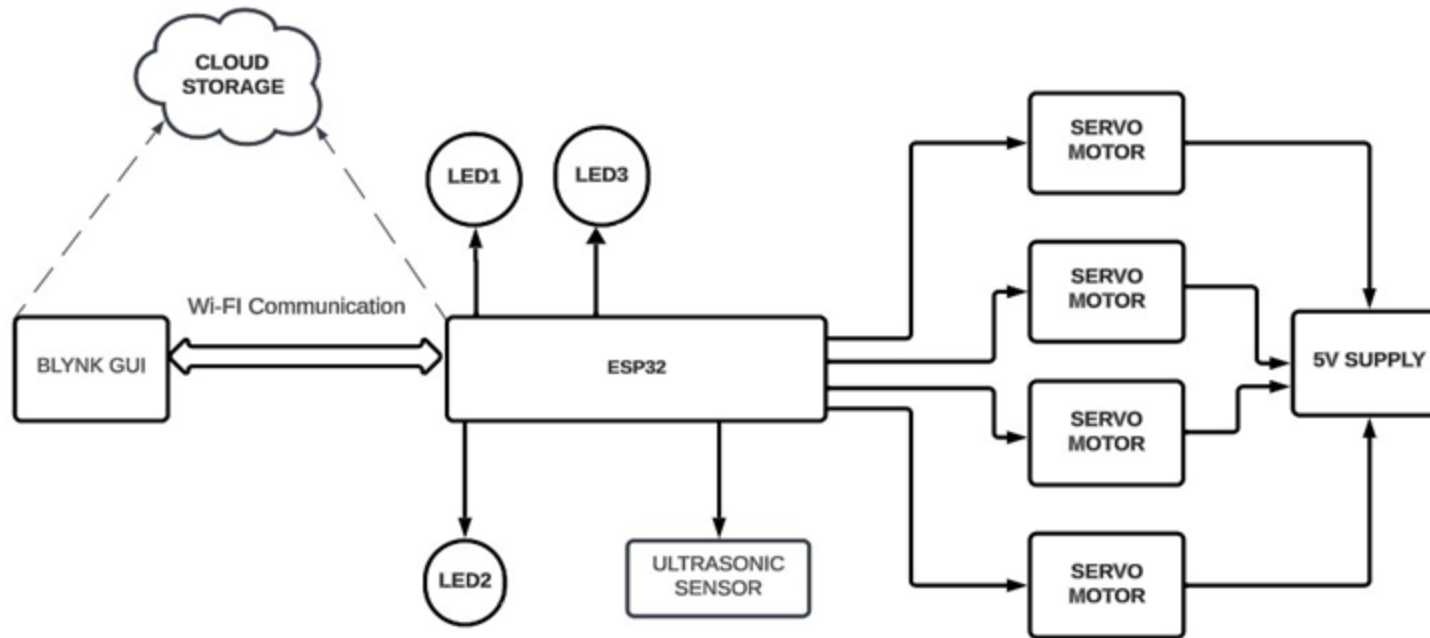
OBJECTIVES AND AIMS

- To design and develop an IoT-based robotic arm that detects, picks, and places construction materials using an ESP32 microcontroller and Blynk application.
- Discover bidirectional communication between control centre and remote unit.

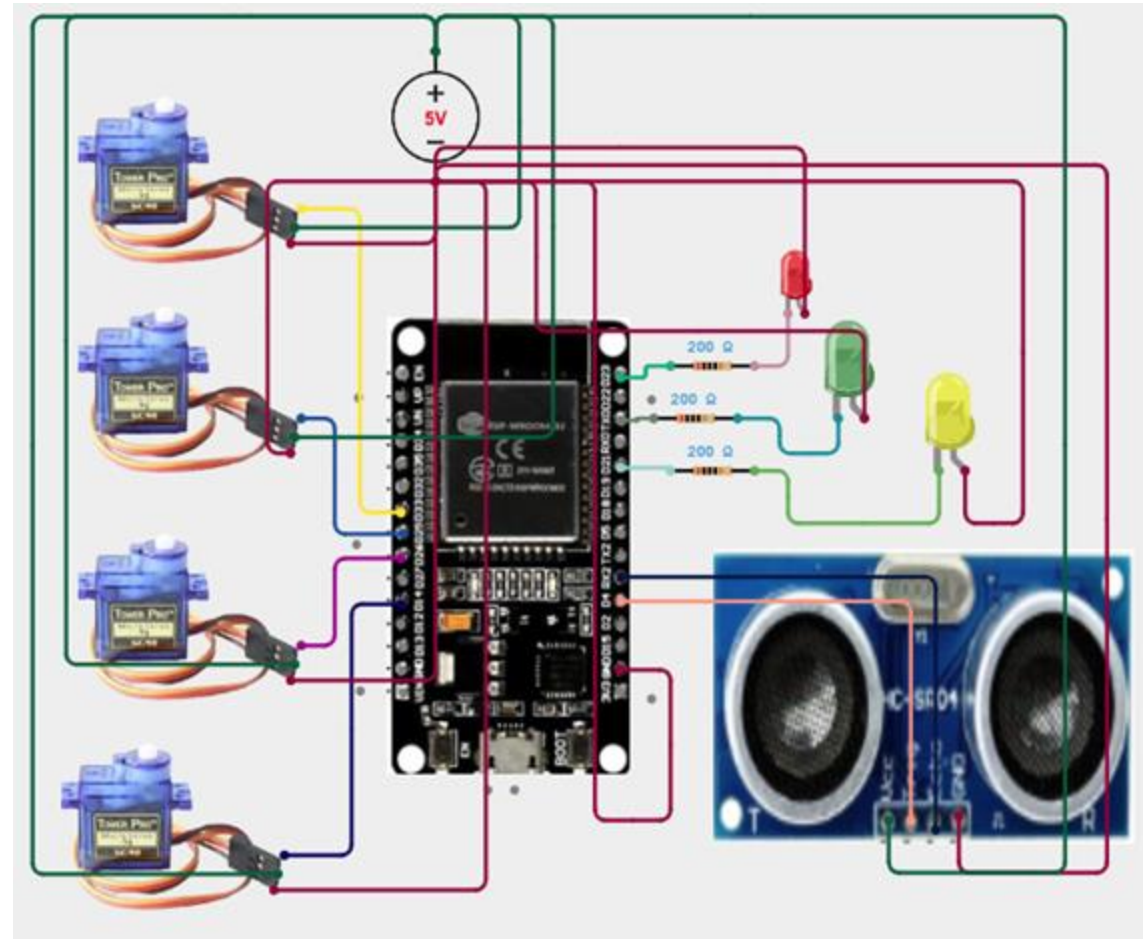
ESPECIFICATINS

5 V to power ESP32 microcontroller to be able to turn on the servo motors.

BLOCK DIAGRAM

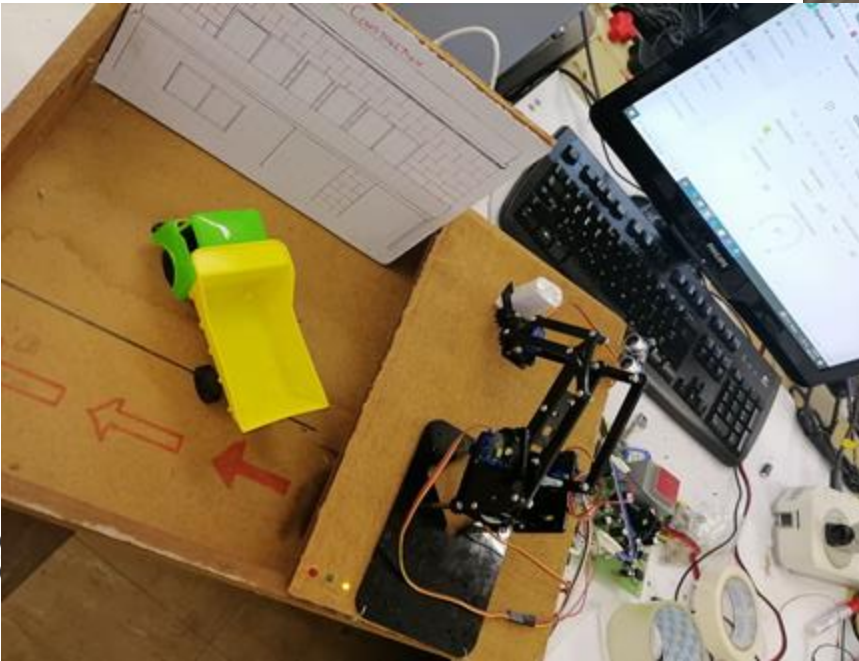


CIRCUIT DIAGRAM



CONSTRUCTION

- Breadboard and component wiring
- LEDs used to indicate manual, automatic and detection mode.
- Blynk dashboard with switches and gauges
- Manua/Auto control setup using Blynk virtual pins
- Closed housing for safety and neatness



CONSTRUCTION SITE ROBOTIC ARM - Offline

by SAMUELLO My organization - 2650AM



Live 1m 5m 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100

MODE SWITCH



GRIPPER SERVO



BASE SERVO



SHOULDER SERVO



ELBOW SERVO



ULTRASONIC DISTANCE

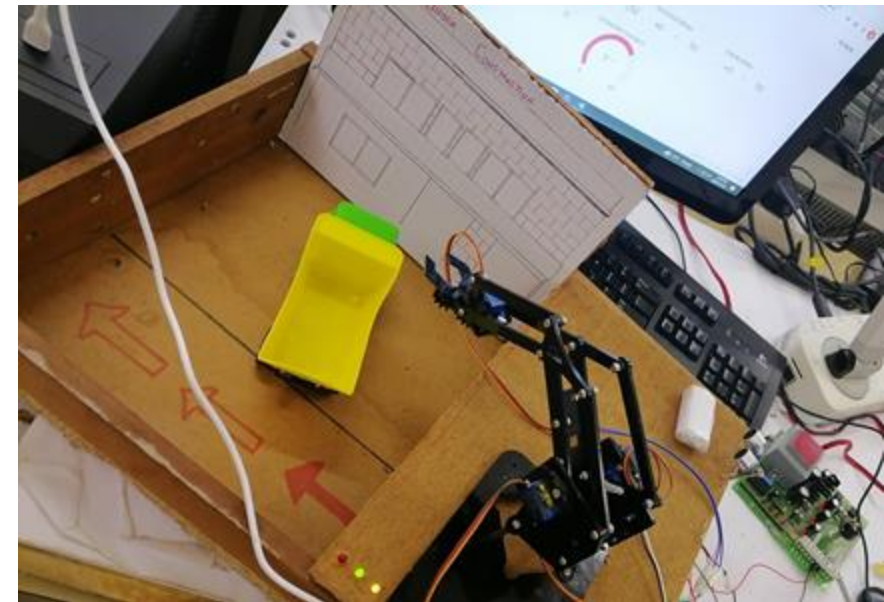
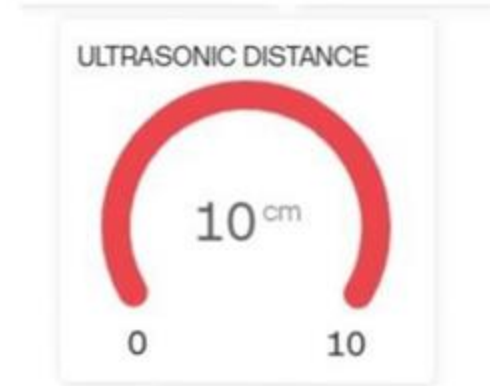


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TESTING

- Sensor readings tested on Blynk gauges (Distance)
- LED indicators tested for Automatic, Manual And Object Detection
- Auto Mode: System tested for automatic response to sensor inputs
- Wi-Fi connection and Blynk communication tested for stability

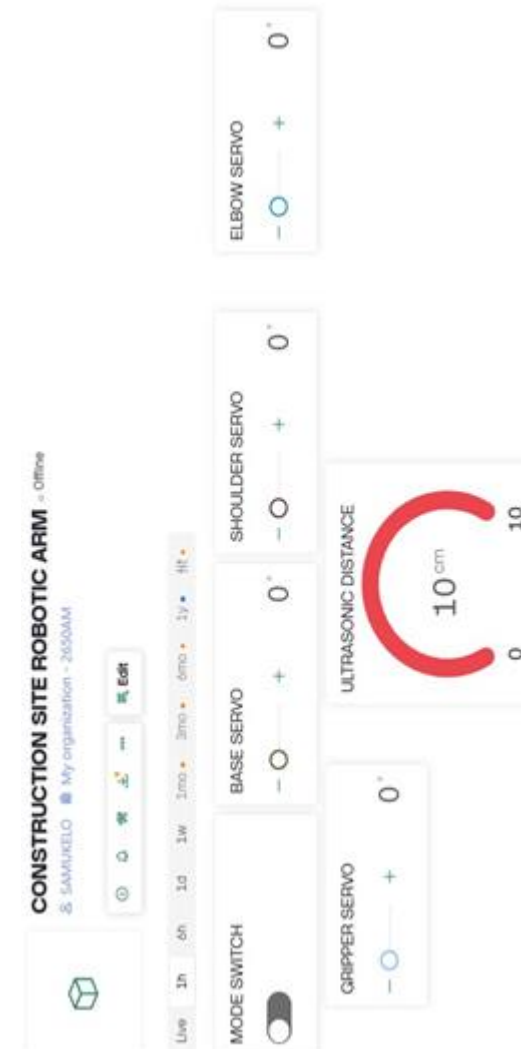


Testing Cont...

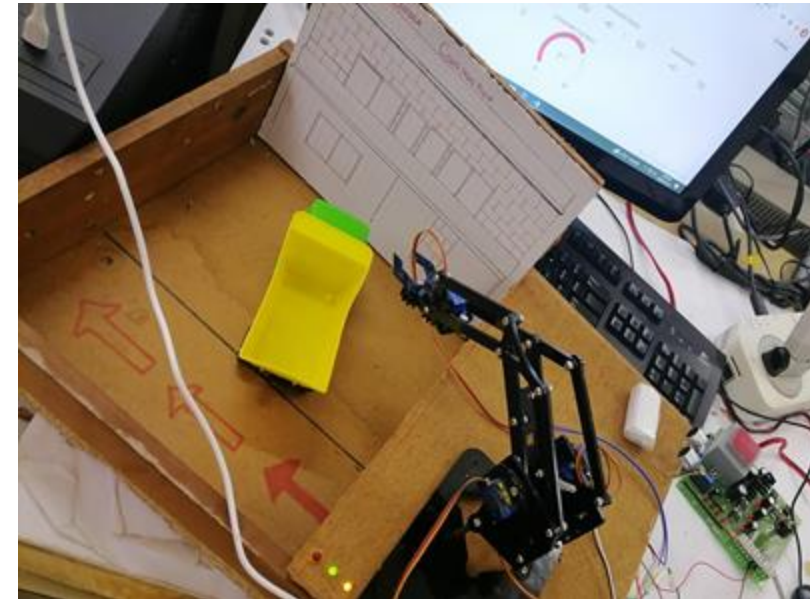
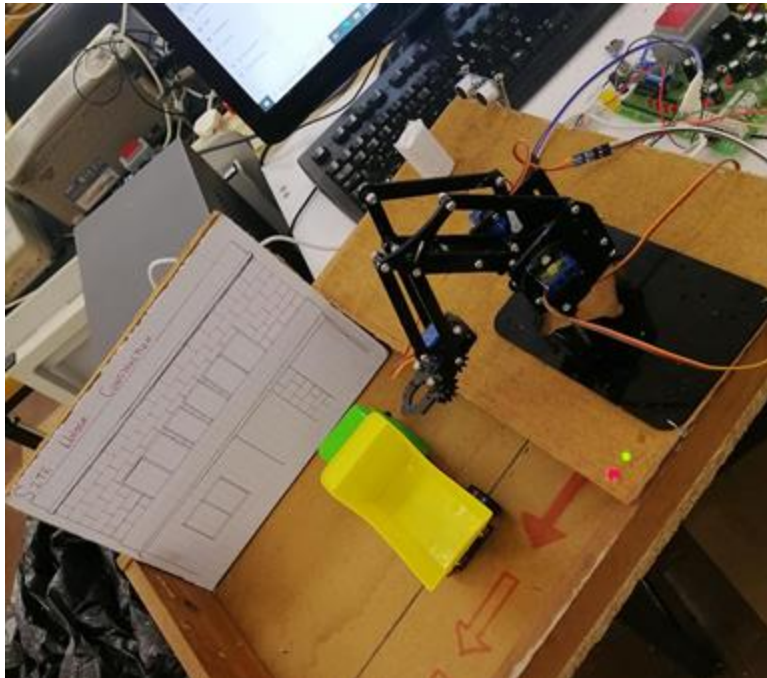
- Manual Mode: All Four Servo Motors were test by controlled via Blynk Virtual switches
- Object detection tested successfully

RESULTS

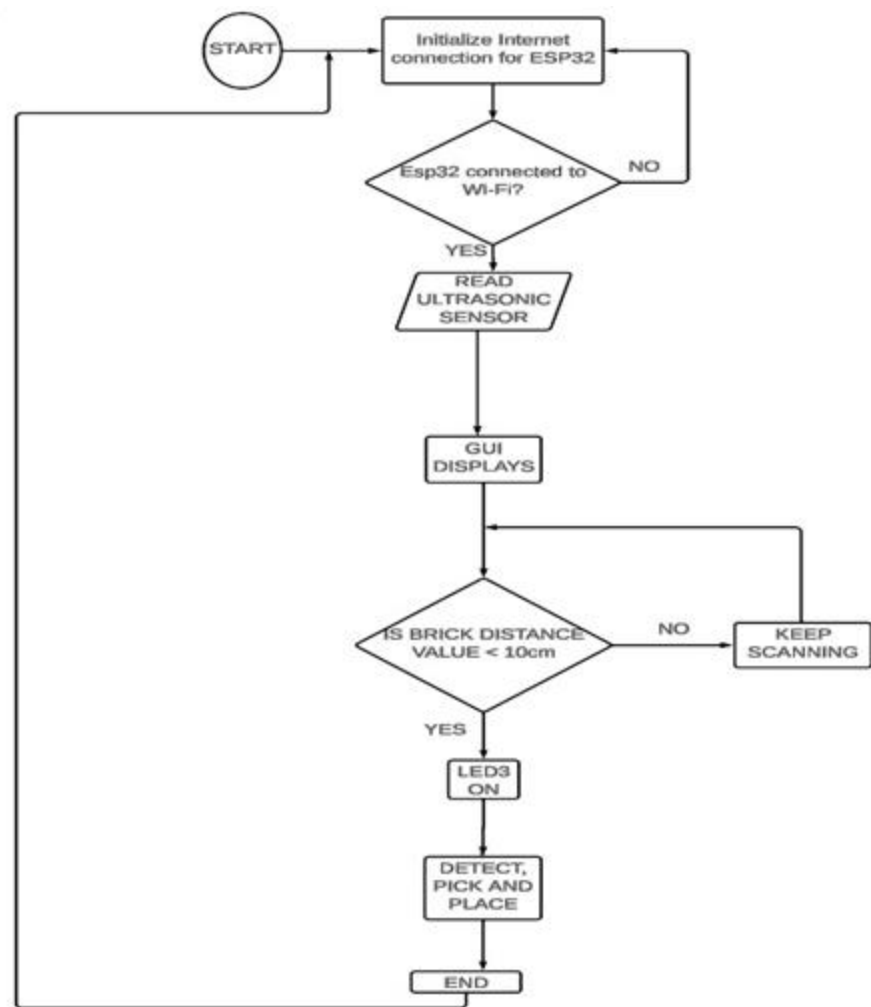
- All Four Servo Motors angles displayed successfully on the Blynk GUI.
- LED1, LED2, LED3 indicated Automatic mode, manual mode and object detected
- Mode switch toggled between Auto and Manual successfully
- Manual Mode: All Four Servo motors controlled successfully via Blynk switches



Results Cont...



FLOW CHART



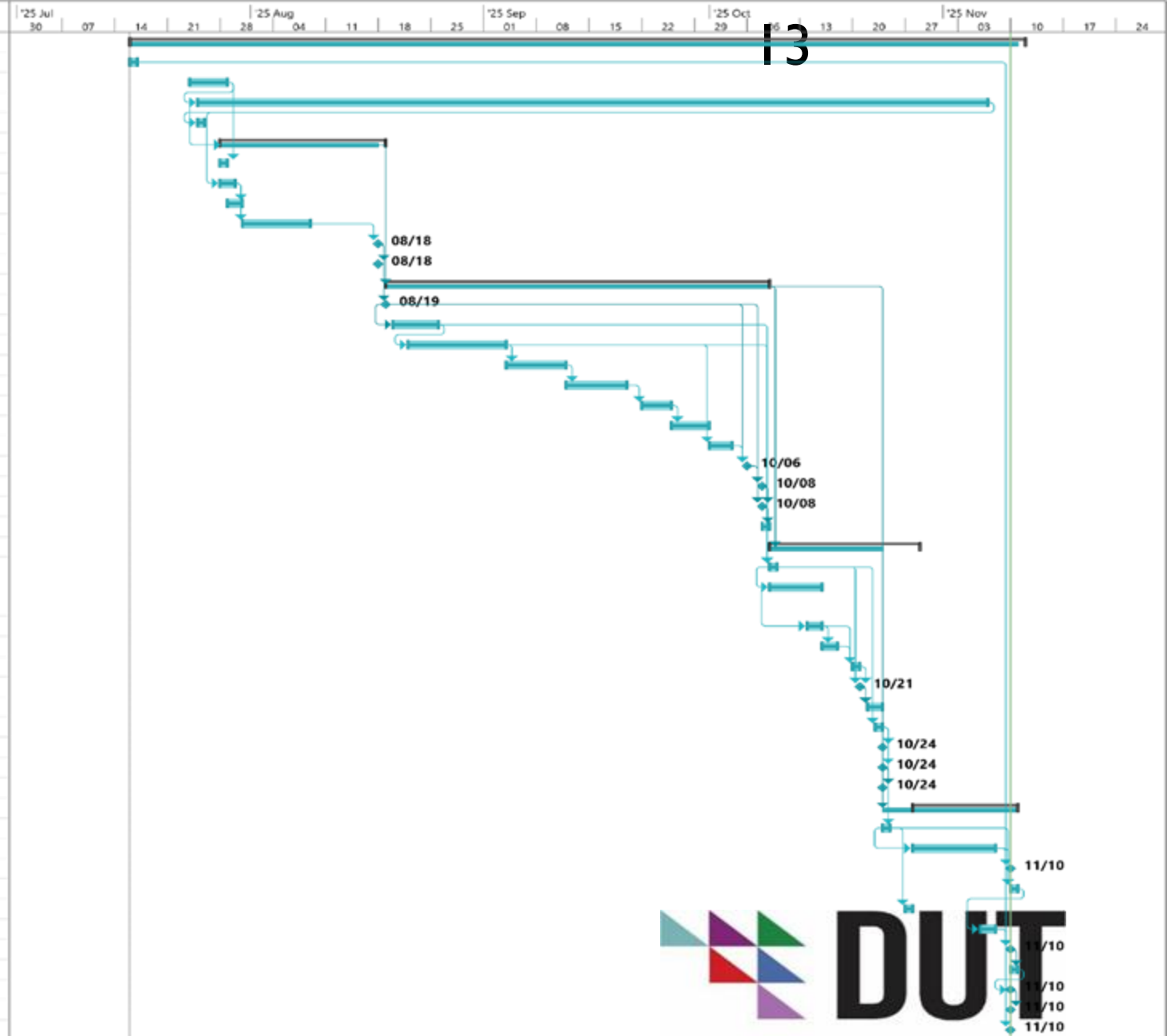
LEESSONS LEANT

- Gained hands-on experience in IoT-based system development.
- Learned to integrate sensors (Ultrasonic sensor and the servo motor) with ESP32 microcontroller
- Developed skills in using Blynk for real-time monitoring and control

COMPLIANCES

- Complied with to instruction of the supervisor of using capacitors used to stabilize the power supply and reduce servo motor shaking.

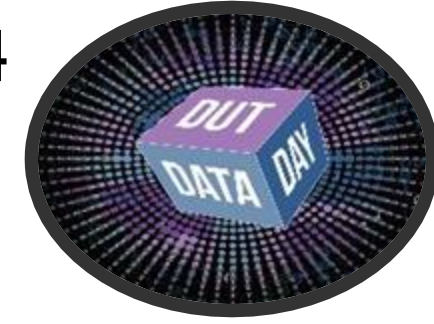
ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1	✓	CONSTRUCTION SITE ROBOTIC ARM	85 days	Wed 25/07/16	Tue 25/11/11	
2	✓	Start Project	1 day	Wed 25/07/16	Wed 25/07/16	
3	✓	Topic allocation	3 days	Thu 25/07/24	Mon 25/07/28	
4	✓	Project research	75 days	Fri 25/07/25	Thu 25/11/06	3
5	✓	Component Search	1 day	Fri 25/07/25	Fri 25/07/25	4
6	✓	Final proposal	16 days	Mon 25/07/28	Mon 25/08/18	455
7	✓	Start proposal	1 day	Mon 25/07/28	Mon 25/07/28	3
8	✓	Research project	2 days	Mon 25/07/28	Tue 25/07/29	4
9	✓	Drawing circuit diagram	2 days	Tue 25/07/29	Wed 25/07/30	8
10	✓	Combining proposal information	7 days	Thu 25/07/31	Fri 25/08/08	8
11	✓	Submission of the proposal	0 days	Mon 25/08/18	Mon 25/08/18	10
12	✓	Proposal Completed	0 days	Mon 25/08/18	Mon 25/08/18	11
13	✓	Project Progress presentation	37 days	Tue 25/08/19	Wed 25/10/08	6
14	✓	Start Progress report	0 days	Tue 25/08/19	Tue 25/08/19	11
15	✓	Step-up BLYNK account	4 days	Wed 25/08/20	Mon 25/08/25	1455
16	✓	Set-up Blynk widget	9 days	Fri 25/08/22	Wed 25/09/03	15
17	✓	ESP32 Testing	6 days	Thu 25/09/04	Thu 25/09/11	16
18	✓	Test wireless communication	6 days	Fri 25/09/12	Fri 25/09/19	17
19	✓	Servo motor testing	4 days	Mon 25/09/22	Thu 25/09/25	18
20	✓	Ultrasonic sensor testing	3 days	Fri 25/09/26	Tue 25/09/30	19
21	✓	Start progress presentation slide	3 days	Wed 25/10/01	Fri 25/10/03	16
22	✓	Progress presentation Submission	0 days	Mon 25/10/06	Mon 25/10/06	14,21
23	✓	Progress Presentation	0 days	Wed 25/10/08	Wed 25/10/08	22
24	✓	Finish progress	0 days	Wed 25/10/08	Wed 25/10/08	14,23
25	✓	Progress Presentation Completed	1 day	Wed 25/10/08	Wed 25/10/08	24
26	✓	Draft report	14 days	Thu 25/10/09	Tue 25/10/28	13
27	✓	Start wireless communication	1 day	Thu 25/10/09	Thu 25/10/09	15,16
28	✓	Wirelss communication implemenation	5 days	Thu 25/10/09	Wed 25/10/15	2755
29	✓	Code ESP32 for Wi-Fi connection	2 days	Tue 25/10/14	Wed 25/10/15	2855
30	✓	Code ESP32 to send Data to BLYNK	2 days	Thu 25/10/16	Fri 25/10/17	29
31	✓	Take results	1 day	Mon 25/10/20	Mon 25/10/20	29,30
32	✓	Finish Wi-Fi communication	0 days	Tue 25/10/21	Tue 25/10/21	27,31
33	✓	Cloud Storage implementation	2 days	Wed 25/10/22	Thu 25/10/23	32
34	✓	Finish wirelss communication	1 day	Thu 25/10/23	Thu 25/10/23	27
35	✓	Prototype Completed	0 days	Fri 25/10/24	Fri 25/10/24	34
36	✓	Draft report submission	0 days	Fri 25/10/24	Fri 25/10/24	34
37	✓	Draft report submitted	0 days	Fri 25/10/24	Fri 25/10/24	36
38	✓	Final report	10 days	Tue 25/10/28	Mon 25/11/10	13
39	✓	Start final report	1 day	Fri 25/10/24	Fri 25/10/24	34
40	✓	Research report	9 days	Tue 25/10/28	Fri 25/11/07	3955
41	✓	Project Housing Completed	0 days	Mon 25/11/10	Mon 25/11/10	40
42	✓	Final report submission	1 day	Mon 25/11/10	Mon 25/11/10	39,40
43	✓	Final report submitted	1 day	Mon 25/10/27	Mon 25/10/27	39
44	✓	Presentation slide	2 days	Thu 25/11/06	Fri 25/11/07	42
45	✓	Presentation slides submission	0 days	Mon 25/11/10	Mon 25/11/10	44
46	✓	Presentation slides submitted	1 day	Mon 25/11/10	Mon 25/11/10	45
47	✓	Project presentation	0 days	Mon 25/11/10	Mon 25/11/10	46
48	✓	Project presentstion completed	0 days	Mon 25/11/10	Mon 25/11/10	47
49	✓	Project Finished	0 days	Mon 25/11/10	Mon 25/11/10	2



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THANK YOU

QUESTIONS ?